

CONTINUOUS DYNAMIC PROGRAMMING

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INTRODUCTION

- Many interesting problems require we move away from discrete state space and action space.
- We will briefly cover the theory behind dynamic programming in such settings, focusing on deterministic cases.
- Exposition and notation follow closely Stokey, Lucas, and Prescott (1989), Chapter 4. See Chapter 9 for the treatment of stochastic dynamic programming.
- Then we will deal with practical issues that arise in dynamic programming.

DETERMINISTIC DYNAMIC PROGRAMMING: THEORY

SEQUENCE PROBLEM

- The problem to be studied in terms of **infinite sequences** is:

$$v^*(x_0) = \sup_{\{x_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t F(x_t, x_{t+1}) \quad \text{s.t. } x_{t+1} \in \Gamma(x_t)$$

- Above, X is the set of possible values for x .
- $\Gamma : X \rightarrow X$ is a correspondence that assigns **feasible values** of the choice variable.
- $F : \text{Gr}(\Gamma) \rightarrow \mathbb{R}$ is a **payoff function**.
- $\beta \in (0, 1)$ is a discount factor.

RECURSIVE PROBLEM: BELLMAN EQUATION

- We focus on the **functional equation**:

$$v(x) = \sup_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}$$

- This is the **Bellman equation**. Its solution is a function v that satisfies this equality for all $x \in X$.
- We will state conditions that ensure $v(x) = v^*(x)$ and the optimal policy of one problem is the same as for the other (the **principle of optimality**)

PRINCIPLE OF OPTIMALITY

- Once we establish the **principle of optimality** we can study properties of the solution to the **functional equation**.
- Define the set of all possible **feasible sequences** for x , given a starting point x_0 :

$$\Pi(x_0) = \{ \{x_t\}_{t=0}^{\infty} \mid x_{t+1} \in \Gamma(x_t) \quad \wedge \quad x_0 \text{ given} \}$$

and $\underline{x} = (x_0, x_1, x_2, \dots)$ is an element.

ASSUMPTIONS

- **Assumption A.1:** Γ is a **nonempty valued** correspondence.
- **Assumption A.2:** For all $x_0 \in X$ and $\underline{x} \in \Pi(x_0)$ the following limit exists (although it might be infinite):

$$\lim_{n \rightarrow \infty} \sum_{t=0}^n \beta^t F(x_t, x_{t+1})$$

- Assumption A.2 holds if F is bounded and $\beta \in (0, 1)$.
- Under assumptions A.1 and A.2, $\Pi(x_0)$ is nonempty valued and problem is well posed. A.1 and A.2 are enough to guarantee that v^* satisfies the functional equation.

PROPOSITIONS ON PRINCIPLE OF OPTIMALITY

Proposition (4.3)

Let X, Γ, F and β satisfy assumption A.1 and A.2, then v^* is a solution to the FE:

$$v^*(x) = \sup_{y \in \Gamma(x)} \{F(x, y) + \beta v^*(y)\}$$

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- For v^* to be the only solution to the FE an extra condition is needed.

Proposition (4.4)

Let X, Γ, F and β satisfy assumption A.1 and A.2, if v is a solution to the FE and for all $x_0 \in X$ and $\underline{x} \in \Pi(x_0)$ it holds that:

$$\lim_{n \rightarrow \infty} \beta^n v(x_n) = 0$$

then $v = v^*$.

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BOUNDED PROBLEMS

- We will establish properties of the solution to the following problem:

$$v(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta v(y)\}$$

$$G(x) = \{y \in \Gamma(x) \mid v(x) = F(x, y) + \beta v(y)\}$$

where v is the **value function** and G is the **policy correspondence**.

- We maintain assumptions A.1 and A.2

ADDITIONAL ASSUMPTIONS

- **Assumption A.3:** X is a **convex subset** of $\mathbb{R}^I$ and Γ is a nonempty, compact valued and continuous correspondence.
- **Assumption A.4:** The function $F : \text{Gr}(\Gamma) \rightarrow \mathbb{R}$ is **bounded and continuous** and $\beta \in (0, 1)$.
- Because F is bounded and continuous we expect the solution to the functional equation lies in the set $C(X)$ of **continuous and bounded functions**.
- What follows is to establish the existence of a solution by means of the **contraction mapping theorem**.
- Define a mapping $T : C(X) \rightarrow C(X)$ as:

$$Tf(x) = \max_{y \in \Gamma(x)} \{F(x, y) + \beta f(y)\}$$

- The solution to the functional equation is then a $v \in C(X)$ such that $v = Tv$.

PROPOSITION 4.7

Proposition (4.7)

Let X, Γ, F and β satisfy assumption A.3 and A.4, and consider $C(X)$ the space of continuous bounded function on X along with the sup norm. Then:

- (a) T defined in (4.5) maps $C(X)$ into itself.
- (b) T defined in (4.5) has a **unique fixed point** $v \in C(X)$, and for all $v_0 \in C(X)$

$$\|T^n v_0 - v\| \leq \beta^n \|v_0 - v\|$$

- (c) Given v the **optimal policy correspondence** $G(x) = \{y \in \Gamma(x) \mid v(x) = F(x, y) + \beta v(y)\}$ is nonempty, compact valued and u.h.c.

PROPERTIES: MONOTONICITY

- **Assumption A.5:** For all y , $F(\cdot, y)$ is strictly increasing in its first l arguments.
- **Assumption A.6:** Γ is monotone in the sense that if $x \leq x'$ the $\Gamma(x) \subseteq \Gamma(x')$.

Proposition (4.8)

Let X, Γ, F and β satisfy assumption A.3 to A.6, and let v be the unique solution to (4.4), then v is *strictly increasing*.

PROPERTIES: CONCAVITY

- **Assumption A.7:** F is strictly concave in both arguments.
- **Assumption A.8:** Γ has a convex graph.

Proposition (4.9)

*Let X, Γ, F and β satisfy assumption A.3, A.4, A.7 and A.8, and let v' be the unique solution to (4.4), then v is **strictly concave** and G is a continuous single valued function.*

DIFFERENTIABILITY

- **Assumption A.9:** F is continuously differentiable on the interior of its domain, $\text{Gr}(A)$.

Proposition (4.10 - Benveniste-Scheinkman)

Let X, Γ, F and β satisfy assumption A.3, A.4, and A.7 to A.9, and let v be the unique solution to (4.4). If $x_0 \in \text{Int}X$ and $g(x_0) \in \text{Int}\Gamma(x_0)$ then v is *continuously differentiable* at x_0 with derivatives given by:

$$v_i(x_0) = F_i(x_0, g(x_0))$$

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- **Proof.** Stokey, Lucas, and Prescott (1989, sec. 4.2, pp. 85).

PRACTICAL ISSUES

A BENCHMARK MODEL

- We use the **neoclassical growth model** as a benchmark to illustrate computational methods.
- The Bellman equation is:

$$v(k, z) = \max_{k' \in \Gamma(k, z)} \left\{ F(k, k', z) + \beta \sum_{z' \in Z} Q(z, z') v(k', z') \right\}$$

where

$$F(k, k', z) = u(zf(k) + (1 - \delta)k - k')$$

$$\Gamma(k, z) = [0, zf(k) + (1 - \delta)k]$$

- Both **state space** $\mathbb{R}_+ \times Z$ and **action space** $\mathbb{R}_+$ are infinite.

ASSUMPTIONS

- $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ is continuous, strictly increasing, strictly concave, C^2 with

$$\lim_{c \rightarrow 0} u'(c) = \infty, \quad \lim_{c \rightarrow \infty} u'(c) = 0$$

- $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ is continuous, strictly increasing, strictly concave, C^2 with

$$\lim_{k \rightarrow 0} f'(k) = \infty, \quad \lim_{k \rightarrow \infty} f'(k) = 0, \quad f(0) = 0$$

- Do these assumptions ensure A.3–A.9 are satisfied?

DISCRETIZATION AND INTERPOLATION

THE CHALLENGE

- A value function with continuous state variables – e.g., $v(k, z)$ – is an **infinite-dimensional object**.
- How do we represent it on a computer? How do we solve for it?
- **A solution:** Discretize k (and z), use theory we covered when we discussed MDP.
- **Problem:** Does it really represent the original problem? Tradeoff: lose accuracy (if grid is coarse) or feasibility problems (if it's too fine).
- **Limits number of continuous state variables** for which it is feasible to solve the problem.

REPRESENTING THE VALUE FUNCTION

- We need a **finite representation** of an infinite-dimensional object.
- We begin by constructing a finite grid for capital:

$$k \in \mathcal{K} = \{k_1, k_2, \dots, k_N\}$$

- At each point on this grid, we store the corresponding value function as a **vector**:

$$\mathbf{v}_j = (v_{j,1}, v_{j,2}, \dots, v_{j,N})^\top, \quad \text{where } v_{j,i} \equiv v_j(k_i)$$

- Note: to simplify the notation we ignored the dependence on z .
- Similarly, the optimal policy can be represented as a vector.

INTERPOLATING THE VALUE FUNCTION

- Suppose we start with the value function represented on a grid: $\mathbf{v}_j$.
- To compute the Bellman operator, we need to evaluate the value function at points **outside the grid**.
- We can use **linear interpolation** to approximate the value function between grid points. Let $\hat{v}_j$ denote this approximation.
- For $k \in [k_{i-1}, k_i]$, we assume that $\hat{v}_j(k)$ is linear and given by:

$$\hat{v}_j(k) = \frac{k - k_{i-1}}{k_i - k_{i-1}} v_{j,i} + \frac{k_i - k}{k_i - k_{i-1}} v_{j,i-1}$$

- This allows off-grid optimization (finding exact maximizers).

ACCURACY OF LINEAR INTERPOLATION

- The accuracy of a linear interpolation depends on how we construct the grid.
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- We can approximate even very non-linear functions, but this may require a very fine grid.
- Quality of approximation may not be **uniform**:
 - Worse approximation near the boundaries
 - Uniform grid can be very inefficient
 - It does not allow to focus where it is needed
- Use **non-uniform grids** with more points where curvature is high.

GRID BOUNDARIES

- **Lower bound:** Do not choose $k_{\min} = 0$.
 - Since $f(0) = 0$ if $\lim_{c \rightarrow 0} u(c) = -\infty$ (which often the case), there will be numerical issues.
 - Choose small positive value instead

- **Upper bound:** Since $\lim_{k \rightarrow \infty} f'(k) = 0$, for each z there exists $\bar{k}_z$ such that

$$zf(k) + (1 - \delta)k < k \quad \text{for all } k > \bar{k}_z$$

- Set $k_{\max} = \max_z \bar{k}_z$ to ensure all feasible capital levels are covered.
- **Number of points:** Trade-off between accuracy and computational cost.

SPACING OF GRID POINTS

- One solution: put more grid points where v has more curvature.
- Another one: put more points near the parts of the function that are more relevant.
- Usually, for low level of assets, capital and so on. In some DP problems, the value function may have a kink or large curvature somewhere else. Recall McCall job search model.
- There are many possible ways of doing it.

EXPANDING GRID

- We can create a grid as follows:
- **Step 1.** First, create an equally-spaced $[0, 1]$ grid:

$$\{\omega_j : \omega_j = \frac{j-1}{N-1}, j = 1, \dots, N\}$$

- **Step 2.** Shift and expand the grid: $k_j = k_{\min} + (k_{\max} - k_{\min})\omega_j^\theta$, where $\theta > 1$ is the expansion factor.
- This creates **denser spacing** near $k_{\min}$ and **sparser spacing** near $k_{\max}$.
- Useful when you want more precision at lower values of state variable.

DOUBLE-EXPONENTIAL GRID

- An alternative is to use a **double-exponential grid** for the state variable.
- Clusters grid points near the lower bound.
- Let ω_j denote the uniform grid on the unit interval $[0, 1]$ defined above.
- Define the grid points as:

$$k_j = k_{\min} + (k_{\max} - k_{\min}) \frac{e^{\alpha \omega_j} - 1}{e^{\alpha} - 1}$$

- The parameter $\alpha > 0$ controls the degree of clustering.
 - As $\alpha \rightarrow 0$, the grid becomes approximately **uniform**
 - For large α , the grid is very **dense** near the lower bound

SOLVING THE MAXIMIZATION STEP

DIRECT MAXIMIZATION

- Recall the Bellman equation:

$$v(k, z) = \max_{k' \in \Gamma(k, z)} \left\{ F(k, k', z) + \beta \sum_{z' \in Z} Q(z, z') v(k', z') \right\}$$

- We can solve the **maximization problem directly** using optimization techniques.
- Important: we do not need to assume $k' \in \mathcal{K}$ (i.e., k' on the grid).
- This gives hope of finding the **exact maximizer** k' .
- Problem:** we do not know $v(k', z')$ at points (k', z') not on grid.

COMPUTATIONAL STRATEGY

- Use an **approximation** $\hat{v}$ of v at off-grid points (e.g. linear interpolation).
- Any optimizer can be used to solve:

$$\max_{k' \in \Gamma(k, z)} \left\{ F(k, k', z) + \beta \sum_{z' \in Z} Q(z, z') \hat{v}(k', z') \right\}$$

- **Computational trick:** Instead of approximating v and summing inside the optimizer, approximate the **expected value function**

$$w(k', z) := \sum_{z' \in Z} Q(z, z') v(k', z')$$

directly, giving $\hat{w}(k', z)$. Maximization becomes $\max_{k'} \{F(\cdot) + \beta \hat{w}(k', z)\}$.

SUMMARY: DIRECT MAXIMIZATION

- Start with grid $\mathcal{K} \times \mathcal{Z}$.
- For each grid point, solve maximization using an **numerical optimizer**.
- Requires approximation $\hat{v}$ (or $\hat{w}$) at off-grid points.
- **Important:** approximation $\hat{v}$ should preserve the shape of v !
- Piecewise linear interpolation preserves concavity if v is concave.

ALTERNATIVE: ROOT-FINDING ON FOCS

- The maximization problem has a first order condition:

$$F_2(k, k', z) + \beta \sum_{z' \in Z} Q(z, z') v_1(k', z') = 0$$

- Instead of solving maximization, solve FOC (a **nonlinear equation**) for k' .
- Note: what guarantees the interior solution?
- Can use **bisection method** or **Newton's method**.
- Again: need to approximate v_1 (marginal value) at off-grid points.
- As before, we can work with $\hat{w}(k', z)$.

ENDOGENOUS GRID METHOD

PROBLEM

- Maximization (root-finding) step is usually computationally expensive.
- We can speed things up by e.g., using Howard Policy Iteration.
- In some cases there is an even faster alternative that allows us to avoid root-finding altogether: the **Endogenous Grid Method** (Carroll, 2006).

CONCEPT

- The first order condition:

$$F_2(k, k', z) + \beta w_1(k', z) = 0$$

where $F_2 = -u'(zf(k) + (1 - \delta)k - k')$.

- Standard methods fix k (state) and search for optimal k' (policy).
- The **Endogenous Grid Method** (Carroll, 2006) **inverts this logic**.
- **Question:** Instead of asking "Given k , what is optimal k' ?", ask "Given that I chose k' , what k must I have started with?".

EGM ALGORITHM

- **Step 1:** Fix a grid for k' (tomorrow's capital) and z .
- **Step 2:** Compute RHS of Euler equation: $Y = \beta w_1(k', z)$.
- **Step 3:** Invert the marginal utility to find implied consumption c :

$$c = (u')^{-1}(Y)$$

- **Step 4:** Use budget constraint to find **endogenous state** k :

$$zf(k) + (1 - \delta)k = c + k'$$

This yields a mapping from grid points k'_j to **endogenous state** points k_j .

- **Step 5:** Interpolate (k_j, k'_j) back to original grid to get policy function.

VERIFYING ACCURACY: EULER EQUATION RESIDUALS

THE IDEA

- How to verify that the solution to the discretized problem is a good approximation?
- The first order condition is:

$$F_2(k, k', z) + \beta \sum_{z' \in Z} Q(z, z') v_1(k', z') = 0$$

where $F_2 = -u'(c)$.

- By the [envelope theorem](#) (Prop. 4.10):

$$v_1(k, z) = F_1(k, k', z) = (zf'(k) + 1 - \delta)u'(c)$$

- Evaluate LHS of FOC at various (k, z) including points **not on the grid**.

INTERPRETATION

- For easier interpretation, define:

$$\mathcal{H}(k, z) := c(k, z) - (u')^{-1} \left(\beta \sum_{z' \in Z} Q(z, z') (z' f'(k') + 1 - \delta) u'(c(k', z')) \right)$$

- Euler equation residuals are:

$$\mathcal{R}(k, z) := \frac{\mathcal{H}(k, z)}{c(k, z)}$$

- **Interpretation:** agents make $\mathcal{R}(k, z) \times 100\%$ mistakes when choosing c .
- Residuals should be small, especially for (k, z) visited frequently.